

The Effects of Anchoring a Fatiguing Forearm Flexion Task to a High vs. Low Rating of Perceived Exertion on Torque and Neuromuscular Responses

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Abstract

Ortega, DG, Housh, TJ, Smith, RW, Arnett, JE, Neltner, TJ, Schmidt, RJ, and Johnson, GO. The effects of anchoring a fatiguing forearm flexion task to a high versus low rating of perceived exertion on torque and neuromuscular responses. *J Strength Cond Res* 38(5): e219–e225, 2024—This study examined the torque and neuromuscular responses following sustained, isometric, forearm flexion tasks anchored to 2 ratings of perceived exertion (RPE). Nine men (mean $\pm$ SD: age = 21.0 $\pm$ 2.4 years; height = 179.5 $\pm$ 5.1 cm; body mass = 79.6 $\pm$ 11.4 kg) completed maximal voluntary isometric contractions (MVIC) before and after sustained, isometric, forearm flexion tasks to failure anchored to RPE = 2 and RPE = 8. The amplitude (AMP) and mean power frequency (MPF) of the electromyographic (EMG) signal were recorded from the biceps brachii. Normalized torque was divided by normalized EMG AMP to calculate neuromuscular efficiency (NME). A dependent *t*-test was used to assess the mean difference for time to task failure (TTF). Repeated-measures analysis of variances was used to compare mean differences for MVIC and normalized neuromuscular parameters. There was no significant difference in TTF between RPE = 2 and RPE = 8 ($p = 0.713$). The MVIC decreased from pretest to posttest at RPE = 2 ($p = 0.009$) and RPE = 8 ($p = 0.003$), and posttest MVIC at RPE = 8 was less than that at RPE = 2 ($p < 0.001$). In addition, NME decreased from pretest to posttest ($p = 0.008$). There was no change in normalized EMG AMP or EMG MPF ($p > 0.05$). The current findings indicated that torque responses were intensity specific, but TTF and neuromuscular responses were not. Furthermore, normalized EMG AMP and EMG MPF remained unchanged but NME decreased, likely due to peripheral fatigue and excitation–contraction coupling failure. Thus, this study provides information regarding the neuromuscular responses and mechanisms of fatigue associated with tasks anchored to RPE, which adds to the foundational understanding of the relationship between resistance exercise and the perception of fatigue.

Key Words: perception, fatigue, biceps brachii, electromyography

Introduction

Performance fatigability is a quantifiable decrease in an objective measure of performance (e.g., maximal voluntary isometric contraction [MVIC]) over time due to alterations in contractile function and muscle activation (8,22). In contrast, perceived fatigability is an increase in the perception of fatigue as changes in homeostasis and an individual's psychological state occur during exercise (8,22). Kluger et al. (22) and Enoka and Duchateau (8) have proposed unified taxonomies of fatigue that include both performance fatigability and perceived fatigability. These 2 interdependent concepts of fatigue and their influence on exercise performance may be examined by selecting appropriate outcome variables (8). Recently, a number of studies (3,18,19,37) have assessed the relationship between performance fatigability and perceived fatigability by anchoring sustained, isometric tasks to ratings of perceived exertion (RPE) using the Omnibus-Resistance Exercise 10-point (OMNI-RES) scale (30) and the RPE Clamp Model of Tucker (40).

Ratings of perceived exertion subjectively describe the effort, strain, discomfort, and fatigue experienced during exercise (31)

and have been used as an alternative to the traditional anchoring schemes based on MVIC or percentages of 1-repetition maximum (1RM) loads to prescribe exercise intensity (10,21). Recent studies (3,18,19,37) have examined the physiological mechanisms that influence fatigue following sustained isometric tasks anchored to RPE by assessing fatigue-induced changes in torque and neuromuscular responses. Specifically, pretest to posttest decreases in MVIC have been used as a global measure of fatigue (3,18,37). The decrease in MVIC may result from fatigue-induced processes occurring at, or distal to, the neuromuscular junction (i.e., peripheral fatigue) as well as proximal to the neuromuscular junction (i.e., central fatigue) (38). Furthermore, the amplitude (AMP) and mean power frequency (MPF) of the electromyographic (EMG) signal have been used as indirect measures to make inferences about the motor unit activation strategies that modulate the decrease in torque production resulting from fatigue (3,18,37). In particular, EMG AMP reflects muscle excitation (41) and EMG MPF is related to motor unit action potential conduction velocity (2). Neuromuscular efficiency (NME = normalized torque $\div$ normalized EMG AMP) is used to describe how the contractile elements of muscle respond to neural excitation and also provides insight into the mechanisms contributing to the fatigue-induced decreases in torque (6).

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Recent studies have examined the effects of joint angle (3), anchor scheme, and sex (36) on the torque and neuromuscular responses following fatiguing forearm flexion tasks anchored to RPE. Specifically, Arnett et al. (3) examined the effects of joint angle on the MVIC and neuromuscular responses following fatiguing forearm flexion tasks anchored to RPE = 8 in men and reported parallel decreases in pretest to posttest MVICs at elbow joint angles of 75°, 100°, and 125°, a decrease in EMG AMP at the elbow joint angle of 125°, and no changes in EMG MPF. Smith et al. (36) examined the effects of anchor scheme (RPE = 7 vs. the torque that corresponded to RPE = 7) and sex on the torque and neuromuscular responses as well as time to task failure (TTF). The results indicated that the RPE task had a longer TTF; however, the decreases in MVIC and EMG AMP were not anchor or sex specific, and EMG MPF did not change following either task (36). No study, however, has examined the torque and neuromuscular responses following fatiguing forearm flexion tasks anchored to multiple RPE values. Keller et al. (18) previously assessed the torque and neuromuscular responses following fatiguing bilateral leg extension tasks at RPE = 1, RPE = 5, and RPE = 8 in men and reported parallel decreases in torque and EMG MPF across the 3 intensities but no changes in EMG AMP. There was, however, an inverse relationship between exercise intensity and TTF (18). Thus, although the effects of joint angle (3), anchor scheme, and sex (36) have been assessed, it remains unclear whether there are differences in the torque and neuromuscular responses when fatiguing forearm flexion tasks are anchored to various RPE values. Therefore, the purpose of this study was to examine the torque and neuromuscular responses following sustained, isometric, forearm flexion tasks anchored to RPE = 2 and RPE = 8 (OMNI-RES scale). Based on the previous studies (3,36), it was hypothesized that (a) RPE = 2 would result in greater TTF than RPE = 8; (b) MVIC would decrease from pretest to posttest with no difference between RPE = 2 and RPE = 8; (c) EMG AMP would decrease from pretest to posttest MVIC with no difference between RPE = 2 and RPE = 8; and (d) EMG MPF would not change from pretest to posttest MVIC for either RPE = 2 or RPE = 8.

Methods

Experimental Approach to the Problem

This study examined the effect of exercise intensity on the torque and neuromuscular responses following sustained, isometric, forearm flexion tasks to failure anchored to RPE = 2 and RPE = 8 (OMNI-RES scale). During each laboratory visit, the subjects performed two 3-second MVICs before and after sustained, isometric, forearm flexion tasks at RPE = 2 or RPE = 8, in randomized order, at a fatiguing elbow joint angle of 100°. The pretest to posttest MVIC and neuromuscular parameters (normalized EMG AMP, EMG MPF, and NME) were compared between the 2 perceptual intensities to determine whether the responses were dependent on the RPE value and to make inferences about the fatigue-induced adjustments in motor unit activation strategies following the fatiguing tasks.

Subjects

An a priori sample size calculation (G*Power version 3.1.9.4, Dusseldorf, Germany) using previously reported performance fatigability data (3) indicated that a power of 0.95 required 6 subjects. Nine men (mean $\pm$ SD: age = 21.0 $\pm$ 2.4 years; height =

179.5 $\pm$ 5.1 cm; body mass = 79.6 $\pm$ 11.4 kg) volunteered to participate in this study. The subjects were recreationally active, which was defined as participating in resistance or aerobic exercise for a minimum of 3 days $\cdot$ wk⁻¹ (27), free of upper-body pathologies that would affect performance, and right hand dominant based on throwing preference (42). The subjects in this study were part of a large multiple independent and dependent variable investigation, but none of the data have been previously published. This study conformed to the Declaration of Helsinki and was approved by the University of Nebraska-Lincoln Institutional Review Board for Human Subjects (IRB Approval #: 20220521909FB), and all subjects signed a written informed consent and completed a health history questionnaire before testing.

Procedures

Time Course of Procedures. The subjects visited the laboratory on 3 separate occasions (orientation session, test visit 1, and test visit 2) separated by 24–312 hours. During the orientation session, demographic information was recorded, and the subjects were familiarized with the standardized warm-up, testing protocol, and standardized OMNI-RES (30) instructions (Table 1). Test visits 1 and 2 included the standardized warm-up, two 3-second pretest MVICs to set a perceptual anchor to RPE = 10, sustained, isometric, forearm flexion tasks to failure anchored to RPE = 2 or RPE = 8, and two 3-second posttest MVICs. The MVICs and sustained tasks were performed with the dominant arm at an elbow joint angle of 100°. Throughout the sustained tasks and MVIC testing, the EMG signal was recorded from the dominant arm's biceps brachii (BB) muscle. The time course of procedures is presented in Table 1.

Omnibus-Resistance Scale Standardized Anchoring Instructions. The anchoring instructions used in this study were originally developed by Gearhart et al. (12) and modified by Smith et al. (35) for use during isometric forearm flexion tasks. To promote the proper use of the OMNI-RES scale, the following standardized anchoring instructions were read to each subject during the orientation session and before each sustained task anchored to RPE = 2 and RPE = 8: “You will be asked to set an anchor point for both the lowest and highest values on the perceived exertion scale. In order to set the lowest anchor, you will be asked to lay quietly without contracting your forearm flexor muscles to familiarize yourself with an RPE of zero. Following this, you will be asked to perform a maximal voluntary isometric contraction to familiarize yourself with an RPE of 10. When instructed to match a perceptual value corresponding to the OMNI-RES scale, perceived exertion should be related to these defined anchors.”

Orientation Session. During the orientation session, the subjects' age, height, and body mass were recorded. The subjects were then oriented to the testing position on an upper-body exercise table (UBXT) with the lateral epicondyle of the humerus of the dominant arm aligned with the lever arm of the isokinetic dynamometer, in accordance with the Cybex II (Cybex II International Inc. Medway, MA) User's Manual. Once positioned, the subjects were familiarized with the OMNI-RES (0–10) RPE scale (30) and read the standardized OMNI-RES instructions (30,35) used during testing. The OMNI-RES (0–10) RPE scale has been shown to be valid and reliable for quantifying perception during resistance exercise (30). The subjects then completed the standardized warm-up, two 3-second isometric forearm flexion MVICs to set a perceptual anchor corresponding to RPE = 10, and brief, sustained, isometric forearm

Table 1**The time course of procedures.***

Orientation session	Test visits 1 and 2
1. Informed consent	1. Standardized warm-up
2. Health history questionnaire	2. Read the standardized anchoring instructions (OMNI-RES scale)
3. Age, height, and body mass recorded	3. Pretest: two, 3-s MVICs at an elbow joint angle of 100°
4. Familiarized with testing procedures	4. Sustained, isometric, forearm flexion tasks anchored to RPE = 2 or RPE = 8 (OMNI-RES scale) performed to task failure at an elbow joint angle of 100°, in a randomized order
5. Read the standardized anchoring instructions (OMNI-RES scale)	5. Posttest: two, 3-s MVICs at an elbow joint angle of 100°
6. Standardized warm-up: four, 3-s, submaximal (50–75% max effort), isometric, forearm flexion muscle actions	
7. Two, 3-s, isometric, forearm flexion MVICs to set a perceptual anchor of RPE = 10	
8. Brief (~1 min) sustained, isometric, forearm flexion tasks anchored to RPE = 2 and RPE = 8	

*OMNI-RES = Omnibus-Resistance; MVIC = maximal voluntary isometric contraction; RPE = ratings of perceived exertion.

flexion tasks anchored to RPE = 2 and RPE = 8 to become familiar with the testing and anchoring procedures.

Test Visits. Before and after each test visit, subjects were instructed to avoid upper-body exercise at least 24 hours before testing. During each test visit, the subjects were positioned on the UBXT and performed the standardized warm-up followed by 1 minute of rest. The subjects were then read the OMNI-RES instructions related to the anchoring procedures. For pretesting, the subjects performed two 3-second forearm flexion MVICs at an elbow joint angle of 100° and were provided with strong verbal encouragement. The subjects then performed the sustained, isometric, forearm flexion tasks anchored to RPE = 2 or RPE = 8 (OMNI-RES scale) at an elbow joint angle of 100°, in a randomized order. During the RPE = 2 and RPE = 8 sustained tasks, the subjects were asked their RPE every 30 and 20 seconds, respectively, to ensure compliance with the prescribed anchors. In addition, to maintain appropriate levels of exertion during the sustained tasks, the subjects were continuously advised to be attentive to sensations of strain, intensity, discomfort, and fatigue (20,30) and to relate the levels of exertion to the previously set anchors of RPE = 0 and RPE = 10. Thus, as fatigue developed, the subjects were free to decrease torque to maintain a constant RPE value. The trials were sustained to task failure, defined as torque reduced to zero. At task failure, the TTF was recorded and posttest MVICs were performed in a manner identical to the pretest MVICs.

Electromyographic and Torque Acquisition. In accordance with the recommendations of the Surface Electromyography for the Non-Invasive Assessment of Muscles (14), pregelled surface EMG electrodes (Ag/AgCl, Accusensor; Lynn Medical, Wixom, MI) were placed in a bipolar arrangement (30-mm center to center) on the BB of the dominant arm during each test visit. The skin was carefully shaved, abraded, and cleaned with alcohol

before electrode positioning. The electrodes were placed over the BB between the medial acromion and the antecubital fossa, at one-third the distance from the antecubital fossa. A reference electrode was placed on the radial styloid process of the forearm.

The raw EMG signals were digitized at 2,000 samples per second using a 12-bit analog-to-digital converter (Model MP150; Biopac Systems, Inc., Goleta, CA). The EMG signals were then stored on a personal computer (Dell Inspiron Dell Inc., Round Rock, TX) for signal processing. Custom programs written with LabVIEW programming software (version 22.3f0, National Instruments, Austin, TX) were used to perform signal processing. The EMG signals were digitally band-pass filtered (fourth-order Butterworth) at 10–500 Hz. A 1-second epoch from the center of the 3-second forearm flexion MVIC with the greatest torque production was used to calculate the EMG AMP (μV_{rms}) and EMG MPF (Hz). The MPF was chosen to reflect the power density spectrum and was calculated as described by Kwatny et al. (24). The torque signals were sampled from the digital torque output of the Cybex II dynamometer and stored on a personal computer (Dell Inspiron Dell Inc., Round Rock, TX) for analyses.

Statistical Analyses

The test-retest reliability for MVIC, EMG AMP, and EMG MPF at RPE = 2 and RPE = 8 were assessed with repeated-measures analyses of variances (ANOVAs) to evaluate systematic error, and a 2,1 model was used to determine the intraclass correlation coefficient (ICC) (43). Four, separate, repeated-measures ANOVAs (time: pretest and posttest) $\times$ 2 fixed factors (RPE: 2 and 8) were used to analyze the mean differences for pretest vs. posttest MVIC, normalized (% pretest MVIC) neuromuscular parameters (EMG AMP and EMG MPF), and NME. A p value of ≤ 0.05 was considered statistically significant. Significant interactions were decomposed with follow-up, post hoc, Bonferroni-corrected, dependent t -tests (43), and a p value

Table 2

Absolute (mean $\pm$ SD) values and mean differences (95% CI) for MVIC torque, EMG AMP, EMG MPF, and NME before (pretest) and after (posttest) the sustained, isometric, forearm flexion tasks to failure anchored to RPE = 2 and RPE = 8.*

	RPE = 2			RPE = 8		
	Pretest	Posttest	Mean difference (95% CI)	Pretest	Posttest	Mean difference (95% CI)
MVIC (Nm)	82.1 $\pm$ 21.0	72.6 $\pm$ 19.8	6.8 to 12.2	80.7 $\pm$ 22.9	59.5 $\pm$ 18.5	16.3 to 26.3
EMG AMP (μV_{rms})	1,237.0 $\pm$ 585.0	1,220.9 $\pm$ 623.8	–10.8 to 43.0	1,256.5 $\pm$ 688.8	1,154.5 $\pm$ 661.3	–4.7 to 208.7
EMG MPF (Hz)	74.1 $\pm$ 12.0	78.6 $\pm$ 12.0	–8.2 to 0.8	83.2 $\pm$ 10.1	80.0 $\pm$ 14.7	–0.4 to 6.8
NME	1.00 $\pm$ 0.00	0.91 $\pm$ 0.59	0.07 to 0.11	1.00 $\pm$ 0.00	0.84 $\pm$ 0.19	0.10 to 0.22

*CI = confidence interval; MVIC = maximal voluntary isometric contraction; EMG AMP = electromyographic amplitude; EMG MPF = electromyographic mean power frequency; NME = neuromuscular efficiency; RPE = ratings of perceived exertion.

Table 3

Reliability data for MVIC torque and neuromuscular parameters (EMG AMP and EMG MPF) during the pretest forearm flexion MVICs at RPE = 2 and RPE = 8.*†

Variables (mean ± SD)	Visit 1	Visit 2	P	ICC	ICC _{95%}
MVIC (Nm)	80.2 ± 22.1	82.6 ± 21.9	0.405	0.935	0.758 to 0.985
EMG AMP (μVrms)	1,131.8 ± 594.2	1,361.8 ± 658.8	0.041†	0.851	0.347 to 0.967
EMG MPF (Hz)	79.6 ± 14.3	77.7 ± 9.2	0.655	0.506	−0.227 to 0.865

*MVIC = maximal voluntary isometric contraction; EMG AMP = electromyographic amplitude; EMG MPF = electromyographic mean power frequency; RPE = ratings of perceived exertion; P = alpha from the ANOVA for systematic error; ICC = intraclass correlation coefficient; ICC_{95%} = ICC 95% confidence interval; ANOVA = analysis of variance.

†Significant mean difference from visit 1 to visit 2.

of ≤ 0.0167 was considered statistically significant. A dependent *t*-test was used to determine the mean difference for TTF. Effect sizes were reported as partial eta squared (η_p^2) for the ANOVAs and Hedge's *g* correction for the pairwise comparisons. Partial eta-squared (η_p^2) effect sizes were interpreted as small (0.01–0.59), moderate (0.06–0.149), and large (≥ 0.15) (5). Hedge's *g* effect sizes were interpreted as trivial (< 0.19), small (0.20–0.59), moderate (0.60–1.19), large (1.20–1.99), very large (2.0–4.0), and extremely large (> 4.0) (15). All statistical analyses were completed in IBM SPSS v. 28 (Armonk, NY).

Results

The pretest and posttest values for MVIC, EMG AMP, EMG MPF, and NME are presented by RPE in Table 2.

Reliability

Table 3 includes the test-retest reliability parameters (*p* value [systematic error], ICC, and ICC_{95%}) for MVIC, EMG AMP, and EMG MPF. There were no mean differences ($p > 0.05$) for test vs. retest for MVIC and EMG MPF; however, there was a significant difference for EMG AMP ($p = 0.041$, ICC = 0.851). The ICC values ranged from 0.506 (EMG MPF) to 0.935 (MVIC).

Time to Task Failure

The results of the dependent *t*-test for TTF indicated no significant mean difference between RPE = 2 and RPE = 8 (mean difference = 59.3 seconds, $p = 0.713$, $g = 0.121$ [trivial]).

Maximal Voluntary Isometric Contraction

The 2-way repeated-measures ANOVA results for MVIC indicated a significant 2-way interaction for RPE $\times$ time ($p = 0.009$, $\eta_p^2 = 0.594$ [large]). Follow-up pairwise comparisons (decomposed by RPE) indicated significant decreases from pretest to posttest at RPE = 2 (mean difference = 9.5 Nm, $p = 0.009$, $g = 1.099$ [moderate]; Bonferroni's corrected $p = 0.0167$) and RPE = 8 (mean difference = 21.3 Nm, $p = 0.003$, $g = 1.348$ [large]; Bonferroni's corrected $p = 0.0167$). Follow-up pairwise comparisons (decomposed by time) indicated that the posttest MVIC values were significantly (mean difference = 13.2 Nm; $p < 0.001$, $g = 2.039$ [very large], Bonferroni's corrected $p = 0.0167$) greater at RPE = 2 than at RPE = 8 (Figure 1).

Electromyographic Amplitude

The results of the 2-way repeated-measures ANOVA for normalized EMG AMP indicated no significant interaction for RPE $\times$ time ($p = 0.575$, $\eta_p^2 = 0.041$ [small]) or main effects for

RPE ($p = 0.575$, $\eta_p^2 = 0.041$ [small]) or time ($p = 0.424$, $\eta_p^2 = 0.082$ [moderate]).

Electromyographic Mean Power Frequency

The 2-way repeated-measures ANOVA results for normalized EMG MPF indicated a significant interaction for RPE $\times$ time ($p = 0.041$, $\eta_p^2 = 0.425$ [large]). Follow-up pairwise comparisons (decomposed by RPE) indicated no significant difference in EMG MPF from pretest to posttest at RPE = 2 (mean difference = -7.2% , $p = 0.249$, $g = -0.395$ [trivial]; Bonferroni's corrected $p = 0.0167$) or RPE = 8 (mean difference = 4.0% , $p = 0.393$, $g = 0.287$ [small]; Bonferroni's corrected $p = 0.0167$). Follow-up pairwise comparisons (decomposed by time) indicated no significant difference (mean difference = 11.2% , $p = 0.041$, $g = 0.772$ [moderate]; Bonferroni's corrected $p = 0.0167$) between RPE anchors for posttest EMG MPF (Figure 2).

Neuromuscular Efficiency

The results of the 2-way repeated-measures ANOVA for NME indicated no significant 2-way interaction for RPE $\times$ time ($p = 0.234$, $\eta_p^2 = 0.172$ [large]) or main effect for RPE ($p = 0.234$, $\eta_p^2 = 0.172$ [large]), but there was a significant main effect for time ($p = 0.008$, $\eta_p^2 = 0.605$ [large]). The main effect for time (collapsed across RPE) indicated that the pretest NME was significantly greater than the posttest NME (mean difference = 0.13, $p = 0.008$, $g = 1.111$ [moderate]; Figure 3).

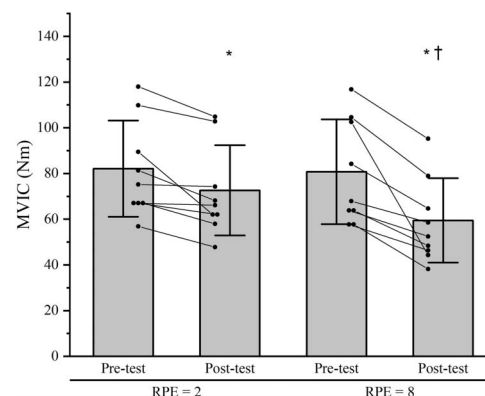


Figure 1. Pretest and posttest (mean ± SD) MVIC values at RPE = 2 and RPE = 8. *Posttest < pretest ($p < 0.05$). †Posttest at RPE = 8 < post-test at RPE = 2 ($p < 0.001$). MVIC = maximal voluntary isometric contraction; RPE = ratings of perceived exertion.

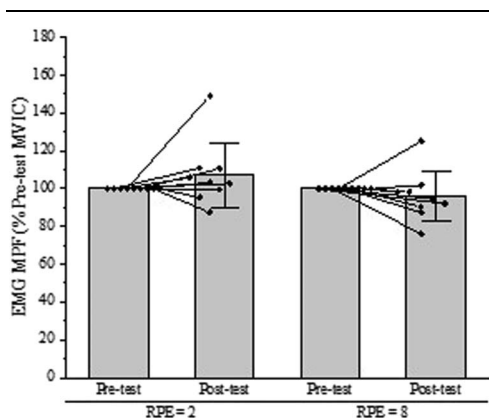


Figure 2. Pretest and posttest (mean $\pm$ SD) normalized EMG MPF values (% pretest maximal voluntary isometric contraction [MVIC]) at RPE = 2 and RPE = 8. No difference between pretest and posttest at RPE = 2 or RPE = 8 ($p > 0.05$). No difference between posttest at RPE = 2 and posttest at RPE = 8 ($p = 0.041$, Bonferroni's corrected $p = 0.0167$). EMG = electromyography; MPF = mean power frequency; RPE = ratings of perceived exertion.

Discussion

The test-retest reliability analyses for the MVIC, EMG AMP, and EMG MPF in this study are presented in Table 3. There were no significant mean differences for the test vs. retest for forearm flexion MVIC and EMG MPF; however, there was a significant ($p = 0.041$) mean difference for EMG AMP. The ICCs for MVIC and EMG AMP were $R = 0.935$ and $R = 0.851$, respectively, and reflected excellent reliability, whereas the ICC for EMG MPF was $R = 0.506$, and reflected fair reliability (4). The degree of variability of the sample can affect the ICCs (23), whereas slight day-to-day changes in the location of the electrodes used to record the EMG signal during the MVICs can affect the absolute values of the neuromuscular parameters (26) used in the test-retest reliability analyses.

In this study, the fatiguing isometric forearm flexion task anchored to RPE = 8 had a greater decrease in MVIC torque (26.2%) than RPE = 2 (11.6%). Previous studies have suggested

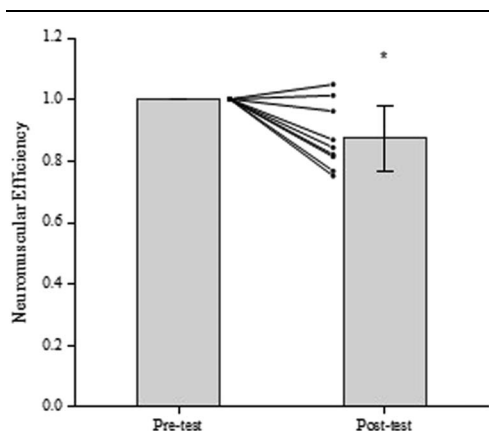


Figure 3. Pretest to posttest (mean $\pm$ SD) neuromuscular efficiency (NME: normalized torque $\div$ normalized EMG AMP values (collapsed across RPE)). *Pretest $>$ posttest ($p = 0.008$). EMG = electromyography; AMP = amplitude; RPE = ratings of perceived exertion.

that for fatiguing tasks anchored to torque (or force), the magnitude of performance fatigability can be affected by the amount of activated muscle mass (32,33,39), the intensity of the exercise bout relative to the critical intensity or maximal load (18,19,39), and the mode (bilateral, unilateral, or whole-body cycling) of the exercise task (18,19,32,33,39). However, less is known regarding the development of fatigue following tasks anchored to RPE (18,19). In this study, the initial torque values for the unilateral, isometric, forearm flexion tasks anchored to RPE = 8 and RPE = 2 were $66.3 \pm 9.6\%$ and $23.3 \pm 9.5\%$ of MVIC, respectively. These values were consistent with differences between the tasks in the amount of active muscle mass and the intensity of the exercise bout. In contrast to the findings of this study, Keller et al. (19) reported no differences in the percent decrease in MVIC torque following sustained, bilateral, isometric leg extension tasks to failure anchored to RPE values above (RPE = 5 and RPE = 8) and below (RPE = 1) the critical force in women. In addition, no differences in the percent decrease in MVIC torque were reported following sustained, bilateral, isometric leg extension tasks to failure anchored to RPE = 1, RPE = 5, and RPE = 8 in men (18). Compared with unilateral forearm flexion tasks, bilateral leg extensions involve a greater amount of muscle mass (7) and torque production (25,28). In theory, differences in engaged muscle mass and torque production could result in differences in muscle oxygenation, adenosine triphosphate utilization, and metabolite buildup during the sustained isometric tasks (13). In addition, it has been hypothesized that fatigue-related differences in TTF between unilateral and bilateral tasks may be due to interhemispheric inhibition (1,18,36). Thus, these findings, in conjunction with those of Keller et al. (18,19), suggested that the magnitude of performance fatigability following fatiguing isometric tasks anchored to RPE was dependent on the exercise mode and muscle group being examined.

The findings of this study indicated no significant difference between RPE anchors for TTF (RPE = 2: 332.3 ± 560.0 seconds vs. RPE = 8: 273.0 ± 188.5 seconds). The present results differed from those of Keller et al. (18) that demonstrated significant intensity-specific differences in TTF following sustained, bilateral, isometric leg extension tasks to failure anchored to RPE = 1, RPE = 5, and RPE = 8 and the corresponding torque values (intensity 1: 433.6 ± 173.6 seconds vs. intensity 5: 149.3 ± 52.4 seconds vs. intensity 8: 95.4 ± 56.9 seconds) in men. Intensity-specific differences in TTF following sustained, bilateral, isometric leg extension tasks to failure were also reported in women (RPE = 1: 470.9 ± 125.4 seconds vs. RPE = 5: 114.8 ± 44.8 seconds vs. RPE = 8: 69.9 ± 30.7 seconds) (19). Previous studies (29,34) have described an inverse relationship between isometric force production and TTF when isometric forearm flexion tasks were anchored to force. Based on that relationship (29,34), it was expected that the RPE = 2 task would result in a longer TTF than the RPE = 8 task. Hunter et al. (16) previously suggested that TTF provides insight into the task-dependent mechanisms contributing to fatigue, which can be dichotomized into perceived fatigability and performance fatigability (22). Therefore, because of task dependency (9), it is possible that the lack of difference in TTF between the RPE = 2 and RPE = 8 tasks in this study was due to the underlying mechanisms modulating perceived fatigability and performance fatigability (8,22) associated specifically with isometric forearm flexion tasks anchored to RPE.

Previous studies (3,36) have reported decreases in EMG AMP (i.e., muscle excitation) from pretest to posttest MVIC following sustained, isometric, forearm flexion tasks anchored to RPE = 7 in a combined sample of men and women (36), as

well as RPE = 8 in men (3). In this study, however, there was no change in EMG AMP from pretest to posttest MVIC at RPE = 2 or RPE = 8. No changes in pretest to posttest EMG AMP were also reported following fatiguing, bilateral, leg extension tasks anchored to RPE = 1, RPE = 5, and RPE = 8 in men (18). Keller et al. (18) suggested that the lack of change in EMG AMP was due to the maintenance of muscle excitation. Decreases in EMG MPF (i.e., motor unit action potential conduction velocity) were also reported following the fatiguing, bilateral, leg-extension tasks (18). In this study, EMG MPF remained unchanged from pretest to posttest MVIC at the high and low perceptual intensities. However, neuromuscular efficiency decreased to a similar extent following each forearm flexion task due to decreases in normalized MVIC torque without a concomitant decrease in normalized EMG AMP. Typically, decreases in NME are representative of peripheral fatigue, which occurs at or distal to the neuromuscular junction (11) when intramuscular metabolites (primarily inorganic phosphate and secondarily hydrogen ions) accumulate (17) and lead to disturbances in excitation-contraction coupling, action potential propagation (37), actin-myosin binding, calcium release and reuptake by the sarcoplasmic reticulum, and troponin-calcium binding (44). Thus, the decreased NME following the sustained, isometric, forearm flexion tasks at RPE = 2 and RPE = 8 likely resulted from peripheral fatigue and excitation-contraction coupling failure. The neuromuscular parameters in this study, however, are indirect measures used to describe fatigue-induced changes in motor unit activation strategies, and, therefore, future studies should use the interpolated twitch technique and potentiated twitch AMP to examine the central and peripheral mechanisms contributing to fatigue during tasks anchored to various RPE values.

The results of this study are limited to male subjects performing sustained, isometric, forearm flexion tasks at a fatiguing elbow joint angle of 100° anchored to RPE = 2 and RPE = 8. Future studies should replicate the results in women to assess potential sex differences in the mechanisms of fatigue, have the sustained tasks performed at fatiguing joint angles above and below the optimal range for torque production to examine the effects of joint angle, and select different RPE values for comparison to examine intensity-specific differences. Furthermore, because of task dependency, future studies should use fatiguing tasks that are dynamic in nature to examine mode-specific differences in the mechanisms of fatigue. In addition, this study sampled the EMG signal from only the BB. Thus, future studies should also sample signals simultaneously from the other forearm flexor muscles (i.e., brachioradialis and brachialis).

In summary, the sustained, isometric, forearm flexion task anchored to RPE = 8 at a fatiguing elbow joint angle of 100° resulted in a greater decrease in torque than that anchored to RPE = 2, despite no difference in TTF. In addition, there were no fatigue-induced changes in EMG AMP or EMG MPF from pretest to posttest MVIC. However, neuromuscular efficiency decreased as a result of both fatiguing tasks, likely due to peripheral mechanisms of fatigue that resulted in excitation-contraction coupling failure. Thus, the current findings indicated that the torque responses following the fatiguing forearm flexion tasks were intensity-specific, but TTF and the neuromuscular responses were not affected by the RPE value used to anchor the tasks.

Practical Applications

This study provides information regarding the neuromuscular responses and mechanisms of fatigue associated with tasks anchored to RPE, which adds to the foundational understanding of the relationship between resistance exercise and the perception of fatigue. The present findings also demonstrated that a sustained task anchored to a high perceptual intensity resulted in a greater pretest to posttest decrease in torque than a low perceptual intensity. Thus, depending on an athlete's current training phase, coaches may have an athlete perform upper-body resistance exercises at a low vs. high RPE value to mitigate the decrease in torque production post training session. In addition, based on individual goals, coaches and practitioners may assess fatigue characteristics of an athlete using exercises anchored to RPE as an alternative to percentages of MVIC or 1RM loads.

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